

Verificación Inicial del Sistema

Para ver la configuración IP se debe aplicar en CMD el comando:
netsh interface ip show config

```
C:\Windows\System32>netsh interface ip show config

Configuration for interface "Ethernet"
    DHCP enabled:                Yes
    IP Address:                  172.21.13.220
    Subnet Prefix:               172.21.0.0/19 (mask 255.255.224.0)
    Default Gateway:             172.21.0.1
    Gateway Metric:               0
    InterfaceMetric:             25
    DNS servers configured through DHCP: 172.16.64.11
                                         172.16.64.12
                                         172.21.64.11
                                         172.21.64.12
    Register with which suffix:   Primary only
    WINS servers configured through DHCP: None

Configuration for interface "Loopback Pseudo-Interface 1"
    DHCP enabled:                No
    IP Address:                  127.0.0.1
    Subnet Prefix:               127.0.0.0/8 (mask 255.0.0.0)
    InterfaceMetric:             75
    Statically Configured DNS Servers: None
    Register with which suffix:   Primary only
    Statically Configured WINS Servers: None
```

Verificación del estado de firewall se debe aplicar en CMD el comando: *netsh advfirewall show allprofiles*

Debe mostrar los perfiles de: Dominio, publico y privado.

```
C:\Windows\system32>netsh advfirewall show allprofiles
```

```
Domain Profile Settings:
```

```
-----  
State ON
```

```
Private Profile Settings:
```

```
-----  
State ON
```

```
Public Profile Settings:
```

```
-----  
State ON
```

Configuración del Firewall

Para activar el firewall se debe aplicar en CMD el comando:
netsh advfirewall set allprofiles state on

```
C:\Windows\system32>netsh advfirewall set allprofiles state on
Ok.
```



Para bloquear trafico entrante ejecutamos el comando:
netsh advfirewall set allprofiles firewallpolicy blockinboundalways,allowoutbound

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.
C:\Windows\system32>netsh advfirewall set allprofiles firewallpolicy blockinboundalways,allowoutbound
Ok.
```

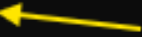
Para bloqueo de Servicios Inseguros (Puerto 445) ejecutamos el comando:

netsh advfirewall firewall add rule name="Bloqueo SMB" dir=in action=block protocol=TCP localport=445

C:\> Administrator: Command Prompt

Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

```
C:\Windows\system32>netsh advfirewall firewall add rule name="Bloqueo SMB" dir=in action=block protocol=tcp localport=445
Ok.
```



Para bloqueo ICMP (Ping) ejecutamos el comando:

netsh advfirewall firewall add rule name="Bloquear Ping" protocol=icmpv4 dir=in action=block

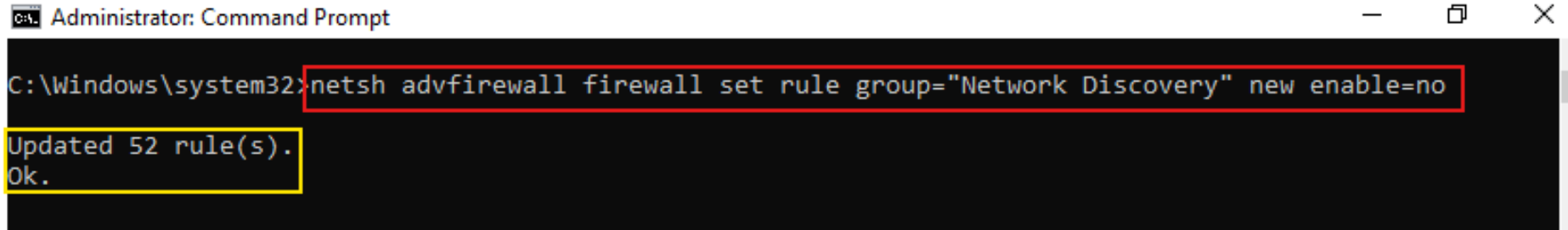
C:\> Administrator: Command Prompt

```
C:\Windows\system32>netsh advfirewall firewall add rule name="Bloqueo Ping" protocol=icmpv4 dir=in action=block
Ok.
```



Para deshabilitar Network Discovery ejecutamos el comando:

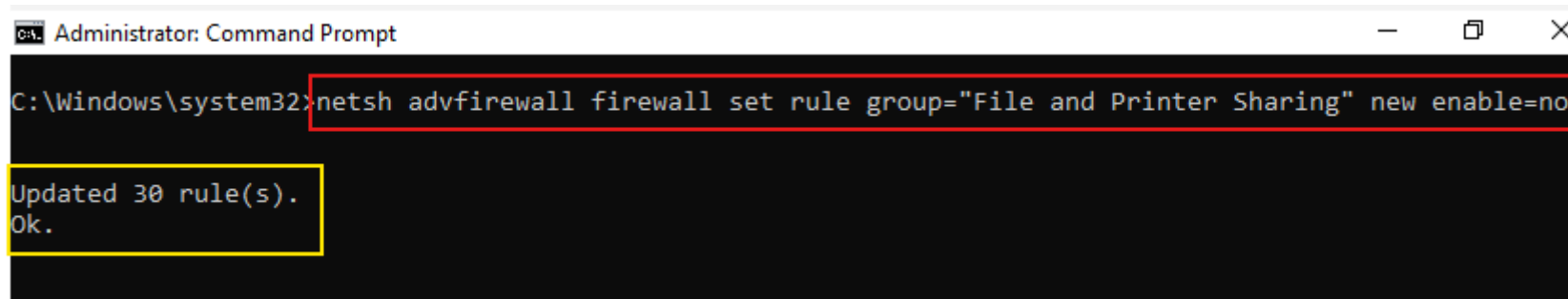
netsh advfirewall firewall set rule group="Network Discovery" new enable=No



A screenshot of an Administrator Command Prompt window. The title bar reads "Administrator: Command Prompt". The command prompt shows the command `netsh advfirewall firewall set rule group="Network Discovery" new enable=no` being entered at the C:\Windows\system32 prompt. Below the command, the output "Updated 52 rule(s).
Ok." is displayed. The command line is highlighted with a red box, and the output is highlighted with a yellow box.

Para deshabilitar File and Printer Sharing ejecutamos el comando:

netsh advfirewall firewall set rule group="File and Printer Sharing" new enable=No



A screenshot of an Administrator Command Prompt window. The title bar reads "Administrator: Command Prompt". The command prompt shows the command `netsh advfirewall firewall set rule group="File and Printer Sharing" new enable=no` being entered at the C:\Windows\system32 prompt. Below the command, the output "Updated 30 rule(s).
Ok." is displayed. The command line is highlighted with a red box, and the output is highlighted with a yellow box.

Protección de Interfaces y Wi-Fi

Para mostrar perfiles Wi-Fi guardados ejecutamos el comando:

netsh wlan show profiles

```
C:\Windows\System32>netsh wlan show profiles
The Wireless AutoConfig Service (wlansvc) is not running.

C:\Windows\System32>
```

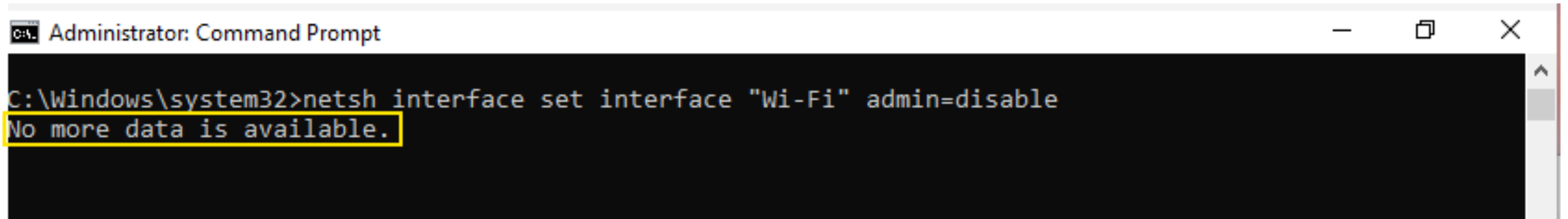
Para Eliminar perfiles Wi-Fi inseguros ejecutamos el comando:

netsh wlan delete profile name="WiFi_Publico"

```
C:\Windows\System32>netsh wlan delete profile name="WiFi_Publico"
The Wireless AutoConfig Service (wlansvc) is not running.
```

Para Deshabilitar Wi-Fi ejecutamos el comando:

netsh interface set interface "Wi-Fi" admin=disable

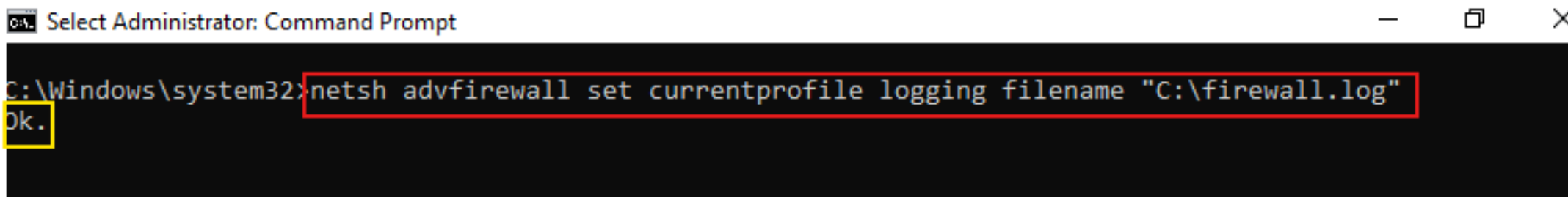


```
Administrator: Command Prompt
C:\Windows\system32>netsh interface set interface "Wi-Fi" admin=disable
No more data is available.
```

Auditoría y Logging

Para habilitar logs del firewall ejecutamos el comando:

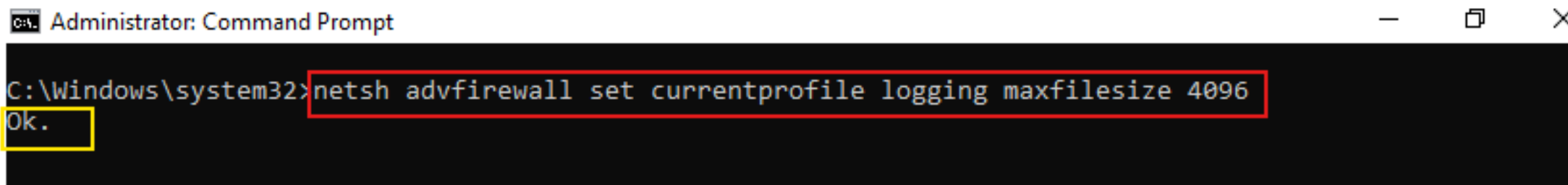
netsh advfirewall set currentprofile logging filename "C:\firewall.log"



```
C:\Windows\system32>netsh advfirewall set currentprofile logging filename "C:\firewall.log"
Ok.
```

Para configurar tamaño máximo del log ejecutamos el comando:

netsh advfirewall set currentprofile logging maxfilesize 4096



```
C:\Windows\system32>netsh advfirewall set currentprofile logging maxfilesize 4096
Ok.
```


Verificación y Auditoría

Para mostrar reglas activas ejecutamos el comando:

netsh advfirewall firewall show rule name=all

```
Administrator: Command Prompt
C:\Windows\system32>netsh advfirewall firewall show rule name=all
```

Rule Name:	Bloqueo Ping	

Enabled:	Yes	
Direction:	In	
Profiles:	Domain,Private,Public	
Grouping:		
LocalIP:	Any	
RemoteIP:	Any	
Protocol:	ICMPv4	
	Type	Code
	Any	Any
Edge traversal:	No	
Action:	Block	

Para mostrar configuración del firewall ejecutamos el comando:

netsh advfirewall show allprofiles

```

C:\Windows\system32>netsh advfirewall show allprofiles

Domain Profile Settings:
-----
State                                ON
Firewall Policy                      BlockInbound,AllowOutbound
LocalFirewallRules                  N/A (GPO-store only)
LocalConSecRules                    N/A (GPO-store only)
InboundUserNotification              Enable
RemoteManagement                    Disable
UnicastResponseToMulticast           Enable

Logging:
LogAllowedConnections                Disable
LogDroppedConnections                Disable
FileName                             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                           4096

```

Para verificar puertos reservados ejecutamos el comando:

netsh interface ipv4 show excludedportrange protocol=tcp

```

C:\Windows\system32>netsh interface ipv4 show excludedportrange protocol=tcp

Protocol tcp Port Exclusion Ranges

Start Port    End Port
-----
5357          5357

* - Administered port exclusions.

```

Pruebas de Conectividad

Ping desde otra máquina (debe fallar):

ping IP_DEL_EQUIPO

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ping 172.21.13.220

Pinging 172.21.13.220 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.21.13.220:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Escaneo de puertos :

nmap IP_DEL_EQUIPO

```
C:\Users\Administrator>nmap -sn 172.21.13.220/19
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-02 18:37 -0400
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
Nmap scan report for 172.21.0.1
Host is up (33s latency).
MAC Address: [REDACTED] (Icann, Iana Department)
Nmap scan report for 172.21.0.4
Host is up (5.4s latency).
MAC Address: [REDACTED] (Cisco Systems)
Nmap scan report for 172.21.0.5
Host is up (4.6s latency).
MAC Address: [REDACTED] (Cisco Systems)
Nmap scan report for 172.21.0.120
Host is up (0.0060s latency).
```

Verificar conectividad saliente (debe funcionar):

ping google.com

```
C:\Users\Administrator>ping -t google.com
```

```
Pinging google.com [142.251.218.46] with 32 bytes of data:
```

```
Reply from 142.251.218.46: bytes=32 time=51ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=80ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=49ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=54ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=68ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=39ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=70ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=83ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=85ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=86ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=74ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=76ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=45ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=37ms TTL=116
```

```
Reply from 142.251.218.46: bytes=32 time=43ms TTL=116
```

Comandos de Auditoría

Para mostrar reglas del firewall ejecutamos:

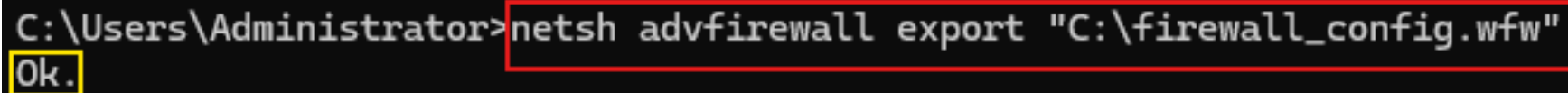
```
netsh advfirewall firewall show rule name=all
```

Para mostrar configuración del firewall ejecutamos:

```
netsh advfirewall show allprofiles
```

Para exportar configuración del firewall ejecutamos:

```
netsh advfirewall export "C:\firewall_config.wfw"
```



A screenshot of a Windows command prompt window. The prompt shows the command `netsh advfirewall export "C:\firewall_config.wfw"` being entered. The command is highlighted with a red rectangular box. Below the command, the output `Ok.` is displayed and highlighted with a yellow rectangular box. The command prompt's title bar and the user's location `C:\Users\Administrator>` are also visible.

```
C:\Users\Administrator>netsh advfirewall export "C:\firewall_config.wfw"  
Ok.
```